Quantum Fault Tolerance Meets Toom's Contours

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1 Hadamard and S Gates

Claim 1.1. For any generator subset S' of size d', denote by $X_{S'}$ and $Z_{S'}$ the following assignments:

$$X_{S'} := \sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S'}} \quad Z_{S'} := \sum_{u\in\langle S/S'\rangle} \theta_{\mathbf{u},\mathbf{S'}}$$

Then $X_{S'}$ and $Z_{S'}$ are binary representations of anticommuting logical codewords of the (d,d')-Toric code. In particular, $X_{S'} \in \mathcal{C}_X/\mathcal{C}_Z^{\perp}$, $Z_{S'} \in \mathcal{C}_Z/\mathcal{C}_X^{\perp}$, and $X_{S'}Z_{S'} = 1$.

Proof. By definition $X_{S'}$ and $Z_{S'}$ are binary assignments on the d'-dimensional faces. We start by showing that $X_{S'} \in \mathcal{C}_X$ and $Z_{S'} \in \mathcal{C}_Z$:

$$\begin{aligned}
\partial^- \sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S'}} &= \sum_{u\in\langle S'\rangle}\sum_{g\in S'} \theta_{\mathbf{u},\mathbf{S'/g}} + \theta_{\mathbf{gu},\mathbf{S'/g}} \\
&= \sum_{g\in S'}\sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S'/g}} + \sum_{u\in\langle S'\rangle} \theta_{\mathbf{gu},\mathbf{S'/g}} \\
&= \sum_{g\in S'}\sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S'/g}} + \sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S'/g}} = 0
\end{aligned}$$

$$\begin{aligned}
\partial^+ \sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S/S'}} &= \sum_{u\in\langle S'\rangle}\sum_{g\in S'} \theta_{\mathbf{u},\mathbf{S'\cup g}} + \theta_{\mathbf{g^{-1}u},\mathbf{S/S'\cup g}} \\
&= \sum_{g\in S'}\sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S'\cup g}} + \sum_{u\in\langle S'\rangle} \theta_{\mathbf{g^{-1}u},\mathbf{S'\cup g}} \\
&= \sum_{g\in S'}\sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S'\cup g}} + \sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S'\cup g}} = 0
\end{aligned}$$

Thus $X_{S'} \in \mathcal{C}_X$ and $Z_{S'} \in \mathcal{C}_Z$, now for showing that they are not binary representations of stabiliers, it's enough to show they admit no vanishing product.

$$X_{S'}Z_{S'} = \sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S'}} \sum_{u\in\langle S/S'\rangle} \theta_{\mathbf{u},\mathbf{S'}} = \theta_{\mathbf{e},\mathbf{S'}}\theta_{\mathbf{e},\mathbf{S'}} = 1$$

□

Definition 1.2 (The Transpose Map)**.** Let d, d' be integers, and let $\mathcal{G}$ be a (d, d')-Toric complex generated by the generator set S. We define the *transpose map* $\phi : \mathcal{F}^{d'} \to \mathcal{F}^{d-d'}$ by acting on the basis elements as follows:

$$\phi(\theta_{\mathbf{v},\mathbf{S'}}) \to \theta_{\phi(\mathbf{v}),\mathbf{S}/\mathbf{S'}}$$

Notice that for $d' = d - d'$, the transpose map sends bits to bits.

$$\phi_{g_a \leftrightarrow g_b}(vu) = \phi_{g_a \leftrightarrow g_b}(\prod_{i \in [d]} (g_i)^{x_i + y_i}) = \prod_{i \in [d]/\{a,b\}} (g_i)^{x_i+y_i} g_a^{x_b+y_b} g_b^{x_a+y_a} = \phi_{g_a \leftrightarrow g_b}(v)\phi_{g_a \leftrightarrow g_b}(u)$$

$$\begin{aligned}
\phi(\theta_{v,S'})_{g_a \leftrightarrow g_b} &= \phi(\bigcup_{I \subset S'} \left\{ \left(\prod_{g \in I} g \right) v \right\})_{g_a \leftrightarrow g_b} \\
&= (\bigcup_{I \subset S'} \left\{ \phi(\left(\prod_{g \in I} g \right) v)_{g_a \leftrightarrow g_b} \right\} \\
&= (\bigcup_{I \subset S'} \left\{ \phi(\left(\prod_{g \in I} g \right) v)_{g_a \leftrightarrow g_b} \right\} \\
&= \theta_{\phi(v)_{g_a \leftrightarrow g_b}, \phi(S')_{g_a \leftrightarrow g_b}}
\end{aligned}$$

So the map sends bits to bits and checks to checks.

Claim 1.3. For $d' = d/2$, the H gate, can be computed transversally.

Proof. The action of the permutation $g_a \leftrightarrow g_b$, $v =$, sends checks to checks.

$$\theta_{\mathbf{v},\mathbf{S'}} \to \theta_{\mathbf{v},\phi(\mathbf{S'})}$$

And:

$$\begin{aligned}
\partial^+ \phi(\theta_{\mathbf{v},\mathbf{S'}}) &= \partial^+ \theta_{\phi(\mathbf{v}),\mathbf{S'}} = \sum_{g \in S/S'} \theta_{\phi(\mathbf{v}),\mathbf{S'} \cup \mathbf{g}} + \sum_{g \in S/S'} \theta_{\mathbf{g^{-1}}\phi(\mathbf{v}),\mathbf{S'} \cup \mathbf{g}} \\
&= 0
\end{aligned}$$

$$\begin{aligned}
\phi(\cdot) &= \phi(\sum_{u \in \langle S' \rangle} \theta_{\mathbf{uv},\mathbf{S'}}) \\
&= \sum_{u \in \langle S' \rangle} \theta_{\phi(\mathbf{uv}),\mathbf{S'}} = \sum_{u \in \langle \phi(S') \rangle} \theta_{\mathbf{u}\phi(\mathbf{v}),\mathbf{S'}}
\end{aligned}$$

$$z\phi(z) = \sum_{u \in \langle S' \rangle} \theta_{\mathbf{uv},\mathbf{S'}} \theta_{\phi^{-1}(\mathbf{uv}),\mathbf{S'}}$$

And there is only one vertex in $\langle S' \rangle v$ for which $\phi(uv) = uv$. But this is not $H^{\otimes n}$. Required condition:

$$|\phi(S) \cap S/S'| \geq 1$$
$$g_1^x g_2^y = g_2^x g_3^y \to x, y = 0$$

It seems that if ϕ is a cycle $g_i \to g_{i+1}$ then it is fine. The prove by induction on $|S'|$, after we throw away the first generator (in the example $g_3 \to y = 0$), then one can assume that $S' \leftarrow S'/g_2$. $\square$

Claim 1.4. The S gate can be computed transversally.

Figure 1: Simulating the S-gate, using the $|S\rangle$ magic state.

2 Toric Encoded Circuits

Definition 2.1 (Troic Encoded Circuit)**.** Let C be a quantum circuit. We say that C is *Toric encoded* if the following holds:

1. There is a Toric code $\mathcal{C}$ such that the qubits of $\mathcal{C}$ can be partitioned into registers $R_1, \ldots, R_k$, and each of the registers is encoded by $\mathcal{C}$.
2. The gates in C with even time coordinates are sweep decoding gates.

For a set of faults $\mathcal{E}$ the gates in $C[\mathcal{E}]$ can be decomposed to cycles, of the following form:

$$C = D_l P_l \Lambda_l, .., D_1 P_1 \Lambda_1$$

Definition 2.2 (Base graph G_o)**.** Let C be a quantum circuit, at depth d, with registers $R_1, \ldots, R_k$, all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by $\mathcal{G}_1, .., \mathcal{G}_k$ the Toric complexes that induce the codes. We define the base graph of C, denoted by $G_o = (V_o, E_o)$ as follows:

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in the union over the vertices in $\mathcal{G}_1, \ldots, \mathcal{G}_k$, and the second coordinate is associated with a time. Namely:

$$V_o := \bigsqcup_i V(\mathcal{G}_i) \times [d]$$

2. The edges E_o are derived from the original edges in $\mathcal{G}_1, \ldots, \mathcal{G}_k$, with the addition that any two vertices in preceding times t and $t+1$ that support qubits (faces) such that C acts non-trivially on those qubits at time t are also connected by an edge. Namely:

$$\begin{aligned} E_o := \big\{\{(v,t),(u,t')\} :\ & \text{if} \\ & \text{either there exists } i \text{ such } \{v,u\} \in \mathcal{G}_i \text{ and } t' = t, t+1 \\ & \text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1}+1\big\} \end{aligned}$$

[COMMENT] Here we should add that a vertex v belongs to location l if there is a qubit f that v belongs to its associated face.

Definition 2.3 (Time Graph of $C(\mathcal{E})$)**.** For a quantum circuit C and set of faults $\mathcal{E}$, let us denote the base graph of $C[\mathcal{E}]$ by G_o. For each time t, denote the deviation induced by $\mathcal{E}$ at time t by $\mathcal{D}_t$, so $\partial\mathcal{D}_t$ is the syndrome that the qubits in the deviation admit. We define the history graph $\mathcal{H} = (V_\mathcal{H}, E_\mathcal{H})$ as follows:

1. The vertices $V_\mathcal{H}$ are taken to be the vertices of G_o, namely tuples $(v,t) \in V_o$, that satisfy that v belongs to a face associated with a check, which, at time t, belongs to the $\partial\mathcal{D}_t$. Namely:

$$\big\{v_t : v_t = (v,t) \in V_o \text{ and there exists a check } f \in C \text{ such } v \in f \in \partial\mathcal{D}_t\big\}$$

2. The edges $E_\mathcal{H}$ is partitioned into two types $A_\mathcal{H}$ and $F_\mathcal{H}$, where $A_\mathcal{H}$ are the edges in E_o restricted to $V_\mathcal{H}$ and $F_\mathcal{H}$ are defined to be the . Namely:

$$\begin{aligned} A_\mathcal{H} &= \big\{e = \{u,v\} \in E_o : u, v \in V_\mathcal{H} \text{ and } T(v) = T(u) \pm 1\big\} \\ F_\mathcal{H} &= \big\{e = \{u,v\} \in E_o :\ \text{exists } w \in V_\mathcal{H} \text{ such } \{v,w\},\{u,w\} \in A_\mathcal{H}\big\} \end{aligned}$$

By definition of the construction, $G_\mathcal{H}$ is a time-graph.

Definition 2.4 (**Size**()-Function)**.** Let S be a subset of vertices, we define the *size* of S to be:

$$\mathbf{Size}(S) := \sum_i^{d+1} \max_{v \in S} L_i(v)$$

Claim 2.5. Let $\mathbf{S} = \{S_1, S_2, .., S_k\}$ where $S_i \subset \mathbb{R}^d$, introduce an edge between S_i and S_j iff $S_i \cap S_j \neq \emptyset$. Assume that $\mathbf{S}$ is connected. Then $\mathbf{Size}(\cup_{i=1}^k S_i) \leq \sum_{i=1}^k \mathbf{Size}(S_i)$.

Proof. It suffices to prove that if $R_1 \cap R_2 \neq \emptyset$, then $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$. Let $a_{i,j} = \max_{v \in R_j} L_i(v)$. Then for any $w \in R_1 \cap R_2$ we have:

$$\begin{aligned} \mathbf{Size}(R_1 \cup R_2) &= \sum_i^{d+1} \max a_{i,1}, a_{i,2} = \sum_i^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\ &\leq \sum_i^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\ &= \mathbf{Size}(R_1) + \mathbf{Size}(R_2) \end{aligned}$$

□

Definition 2.6 (Size Invariant Action)**.** **[COMMENT]** Here we classify both transversal gates, and rotation on the Toric. An action g is be said size invariant action if, for any assignment z, it holds that $\mathbf{Size}(gz) = \mathbf{Size}(z)$.

Claim 2.7. The sweep rule is Size ξ-shrinking.

Proof. $\square$

[COMMENT] Eventually, we might not used the projected graph, and then we will split to history of the slices lemma for 3 parts. The first is when time $= 3t$, in that we apply an invariant action so.